Azure Local Disaster Recovery

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Agenda

- **Objective:** Explore Microsoft-supported Disaster recovery solutions for Azure Local
  Scope: Azure Site Recovery and Hyper-V Replica

Azure Local Resiliency Foundations

- Azure Local VMs are **highly available**, automatically restarting following a node failure on one of remaining nodes
- Well-architected workloads add **redundant app instances** and **multi-tier resiliency**
- DR planning extends beyond HA to address **site-wide failure**, **data corruption**, or **malicious deletion**
- Backups allow **point-in-time recovery**, but they are too slow for critical workloads
- **Continuous replication** facilitates rapid failover using copies of VM disk content

Disaster Recovery on Azure Local

- Protection against failures beyond built-in **platform high availability**
- Combines **replication**, **failover**, **failback**, and **recovery orchestration**
- Two supported technologies:
  - Azure Site Recovery (ASR)
  - Hyper-V Replica
- **ASR** is the primary cloud-based DR mechanism
  - orchestrated, asynchronous replication to Azure,
  - replication frequency of 30s
- **Hyper-V Replica** offers an on-premises alternative
  - asynchronous VM replication between clusters
  - replication frequency can be set to 30s, 5min, or 15min

Introducing Azure Site Recovery (ASR)

- **ASR** replicates Azure Local VMs to Azure for full cloud-based DR.
- Supports **continuous change tracking**, **automated deployment**, **test failover**, and orchestrated **failover/failback**.
- ASR is Azure's primary DR platform for **Azure Local**.

ASR Deployment Models: Automated vs Manual

- **Automated deployment** via ASR extension for Azure Local (preview):
  - Detects cluster nodes automatically
  - Installs ASR agent
  - Applies replication policy
- **Manual deployment** (production-ready):
  - Use **Hyper-V to Azure** disaster recovery configuration
  - Requires explicit configuration on each node

ASR Replication Characteristics

- Supports **frequent replication cycles** with **RPO as low as 30 seconds**.
- Continuously transmits changes from Azure Local VM disks to Azure Storage.
- Replication jobs visible in vault: **health**, **job status**, **protected disks**.

ASR Failover Types

- **Test Failover**:
  - Create an isolated VM in Azure without impacting production.
- **Planned Failover**:
  - Graceful shutdown ensures **zero data loss**.
- **Unplanned Failover**:
  - Triggered during outages; VM brought up even if source is unavailable.
- Post-failover workloads run fully as **Azure VMs**.

ASR Failback

- ASR can reverse replication back to Azure Local after recovery.
- Failback returns workloads to on-premises with Azure-generated changes applied.
- Requirements:
  - Azure Local cluster must be **healthy**
  - If failing back to alternate cluster, VM becomes **unmanaged**
  - Same-cluster failback maintains **Azure Arc connection** if within **45-day window**.

ASR Prerequisites & Requirements

- VMs must be **highly available** to enable replication.
- Nodes must have **internet access** for replication to Azure.
- Administrator requires the **Owner role** on the Recovery Services vault.

ASR Step 1: Prepare Infrastructure (Portal Workflow)

- Initiate the ASR setup process from the Azure portal:
  1. Open **Capabilities → Disaster Recovery**.
  2. Select **Protect VM workloads**.
  3. Choose **Prepare infrastructure**.
- Follow the guided setup to create the required Azure resources:
  - Recovery Services vault
  - Hyper-V Site
  - Replication Policy
- The setup automatically deploys the ASR agent on each node.

ASR Step 2: Enable Replication

- Enable replication via **Recovery Services vault → Replicate**.
- Select **Hyper-V machines to Azure**.
- Configure:
  - **Source environment** (local Hyper-V site)
  - **Target environment** (subscription, resource group, network, storage)
  - **VM selection**
  - **OS/Disks**
  - **Replication policy**
- **Confirm and start replication.**

Monitor ASR Replication

- Track **Replication health**, **Status**, **Job progress**, and **Job IDs**.
- Verify that **OS disk** and **data disks** are marked as **Protected**.
- Wait until replication completed successfully before testing failover.

ASR Step 3: Test Failover

- Configure Azure networking for test failover.
- Run **Test Failover** for individual VMs or Recovery Plans.
- Validate:
  - VM boot
  - Application functionality
  - Network routing & DNS
- Test Failover is, by default, isolated so it does **not** impact production.

ASR Step 4: Recovery Plans

- **Recovery Plans** orchestrate multi-VM failover sequences.
- Allow grouping of VMs into tiers
  - e.g., Domain Controllers → Databases → App Servers
- Support **scripts**, automated steps, and ordered boot.
- Support all **failover** types

ASR Step 5 & 6: Failover and Failback

- **Failover to Azure** completes VM startup using replicated data.
- **Failback** reverses replication and returns workloads on-premises.
- Evaluate network, storage, authentication before failback.

ASR Caveats & Known Issues

- Guest Configuration policies do not run while VM is in Azure.
- Extensions installed by Arc are not manageable inside Azure.
- VM Guest Agent installation can conflict with Arc.
- Installation fails:
  - Hyper-V missing
  - Wrong Hyper-V site association
  - Application Control (WDAC) enforced (use Audit temporarily)

Hyper-V Replica Overview

- Built-in feature providing **asynchronous replication** between two clusters.
- Replicates: VM configuration, VHDs, delta logs.
- Supports:
  - Test failover
  - Planned failover
  - Unplanned failover
  - Reverse replication
  - Failback.

Hyper-V Replica Deployment & Configuration

- **Manual configuration** only (not available via Azure portal).
- Deployment steps:
  - Configure clusters as **replica servers**
  - Set **authentication** (Kerberos or certificates)
  - Configure **firewall rules**
  - Enable replication **per VM**
  - Choose **replication frequency**: 30s / 5m / 15m
- Requires **Hyper-V Replica Broker** role for cluster-to-cluster replication.

Hyper-V Replica Failover, Failback & Behavior

- Failover types:
  - Test
  - Planned
  - Unplanned
- After failover, VM runs as **unmanaged**
- Failback reverses replication direction and restores Azure management if returning to original cluster.
- With alternate-cluster failback, VM stays unmanaged.

Hyper-V Replica Network & Performance Considerations

- Replication consumes **bandwidth** and **IOPS**.
- Lower intervals (e.g., 30 seconds) translate into higher load.
- Requires adequate storage performance at replica site.
- Refer to Hyper-V Replica optimization guidance.

ASR vs Hyper-V Replica

Area	Azure Site Recovery (ASR)	Hyper-V Replica (HVR)
Replication Path	Azure Local → Azure	Azure Local → Azure Local
Target VM Location	Azure VMs	Azure Local VMs
Management Model	Orchestrated, cloud-based	Manual, infrastructure-dependent
Cost Model	Azure usage costs apply	No Azure costs
Failback & Registration	Same cluster (Azure Local VM) or different (unmanaged VM)	Same cluster (Azure Local VM) or different (unmanaged VM)

Combined DR Strategy (ASR + Hyper-V Replica)

- Use **ASR** for:
  - Remote offices
  - Workloads needing cloud-based DR
  - Mission-critical apps requiring automation
- Use **Hyper-V Replica** for:
  - Larger campuses
  - Low-latency on-prem DR
  - Sensitive data that cannot leave site
- Multi-layered DR provides best resilience.

Workload-Specific DR: SQL Server

- SQL Server DR approaches include:
  - Always On Availability Groups (AG)
  - Failover Cluster Instances (FCI)
  - Log Shipping
  - Replication
  - Backups
- Combine SQL-native DR with **host-level DR** for best coverage.

Workload-Specific DR: Azure Virtual Desktop (AVD)

- AVD host pools must be DR-ready:
  - Pooled pools → replicate image, FSLogix profiles
  - Personal pools → full VM replication needed
- DR may use:
  - ASR to Azure
  - Hyper-V Replica to secondary cluster

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